CPS School Analysis Project Report

Neighborhood Disadvantage and School Outcomes in Chicago

PPHA-30538 Final Project

Project Information

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Lecture Section: PPHA-30538, M/W 10.30am/9.00am

Research Question

Chicago is one of the most residentially segregated cities in the U.S., and that shows up in education. Our research question is: **Does neighborhood disadvantage weaken the transition from high school graduation to college enrollment in Chicago?** Our goal is not to make strong causal claims, but to describe patterns and quantify how strongly place is associated with post-secondary opportunities.

Data and Key Variables

We combine three data sources:

1. **American Community Survey (ACS)** – tract-level data for Cook County. We focus on five indicators of neighborhood disadvantage: poverty rate, (log) median income, unemployment rate, share of single-parent households, and rent-burden rate.
2. **CPS School Progress Reports** – school-level outcomes for 128 high schools, including four-year graduation rate, college enrollment rate, and the **transition gap** (graduation minus college enrollment).
3. **Chicago census-tract boundaries** – a KML shapefile from the Chicago Data Portal that provides the geometry for each tract so that we can match schools to neighborhoods.

Approach and Coding Overview

Our analysis pipeline has four main steps, implemented in separate Python scripts:

1. **Clean and join data** (`preprocessing.py`): read raw ACS, CPS, and shapefile data, filter out tracts with implausible values, and assign each school to a census tract.
2. **Build a Neighborhood Disadvantage Index (NDI)** (`preprocessing.py`): because the ACS variables are highly correlated, we use Principal Components Analysis (PCA) to combine the five indicators into a single standardized index for each tract.
3. **Attach NDI to schools and explore patterns** (`eda.py`, `spatial.py`, `plot_spatial.py`): merge the school data with tract-level NDI, create summary tables by NDI quintile, and generate the static plots used in this write-up.
4. **Model relationships and build the dashboard** (`regression_analysis.py`, `app.py`): run linear regressions of outcomes on NDI and build a Streamlit app that allows users to interactively explore the maps and regression results.

Exploratory and Spatial Analysis

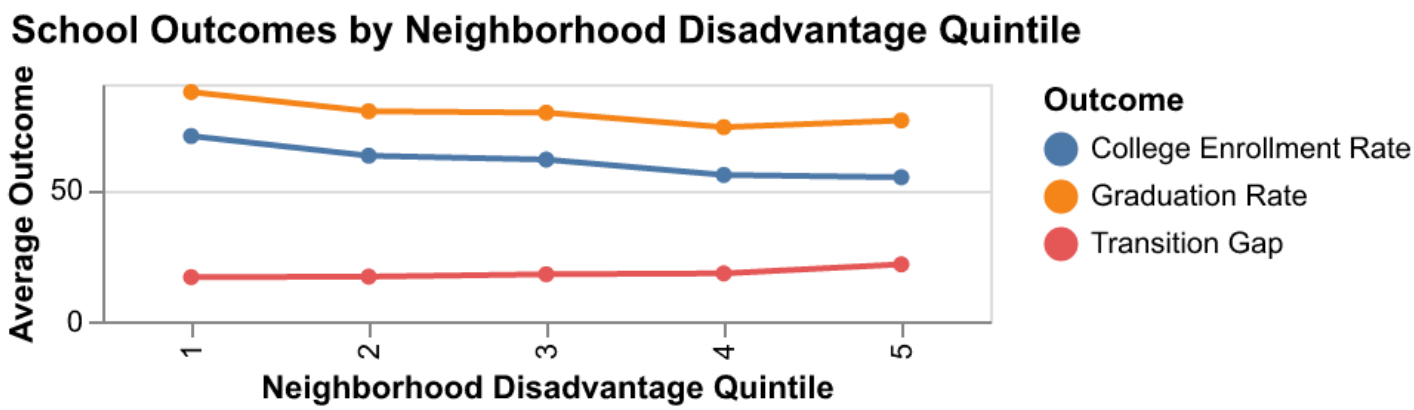
Outcomes by Neighborhood Disadvantage Quintile

To get an initial sense of patterns, we group schools into five quintiles of neighborhood disadvantage, where 1 is the least disadvantaged and 5 is the most disadvantaged. For each quintile we compute the average graduation rate, college enrollment rate, and transition gap.

In the **least disadvantaged neighborhoods (quintile 1)**, the average graduation rate is about **87%**, and the college enrollment rate is about **70%**, leaving a transition gap of roughly **17 percentage points**.

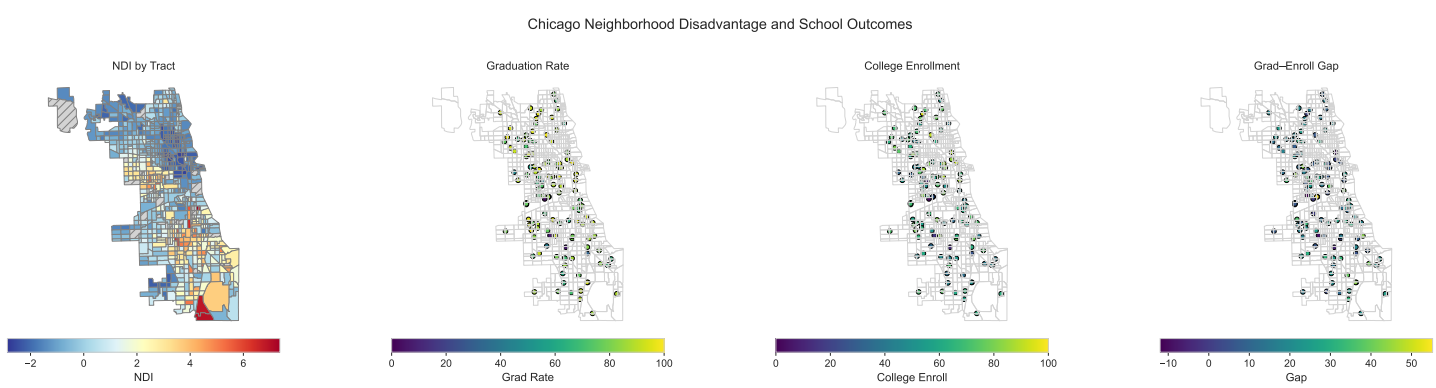
In more disadvantaged areas, both outcomes deteriorate. In the **most disadvantaged neighborhoods (quintile 5)**, graduation recovers slightly, but enrollment remains low, and the transition gap widens to more than **21 percentage points**.

These descriptive patterns already suggest that **place matters**: schools located in more disadvantaged neighborhoods tend to have lower graduation and college enrollment, and a larger gap between the two.



Spatial Patterns in Static Plots

Our static plots make these patterns visible on the map:
1. A **choropleth map of the Neighborhood Disadvantage Index** shows that lower disadvantage is concentrated on the North Side and along the lakefront, while higher disadvantage clusters on the South and West Sides.
2. A **map of schools colored by graduation rate** suggests that graduation has improved city-wide: high-graduation schools are scattered across both advantaged and disadvantaged neighborhoods.
3. A **map colored by college enrollment rate** reveals stronger spatial inequality: schools in more advantaged neighborhoods show much higher college enrollment than many schools on the South and West Sides.
4. A **map colored by the transition gap** highlights where students are most likely to be "lost" between graduation and enrollment; larger gaps tend to appear in more disadvantaged areas.
Together, these tables and maps provide a descriptive, non-causal answer to our research question: neighborhood disadvantage is closely associated with whether graduation translates into college enrollment.

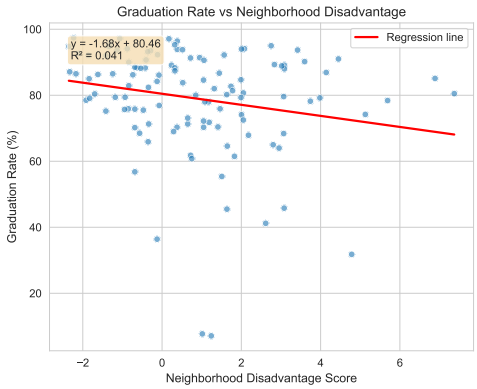


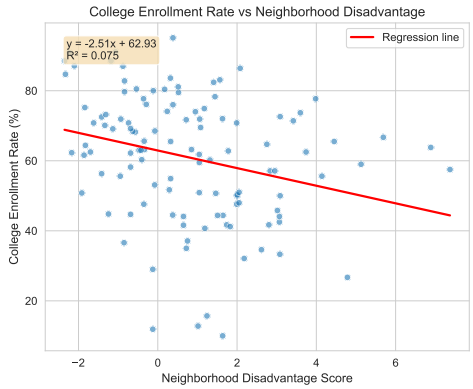
Regression Results and Interpretation

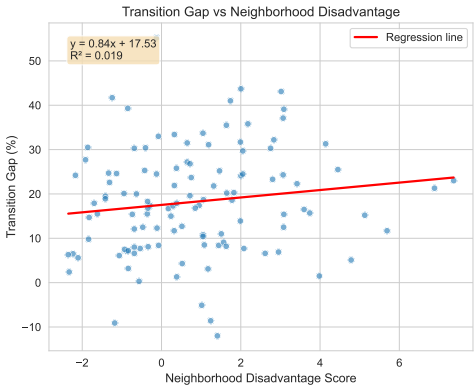
We estimate three simple linear regressions at the school level, using NDI as the main predictor and separately modeling (1) graduation rate, (2) college enrollment rate, and (3) the transition gap.

The results show a clear pattern. A one-unit increase in neighborhood disadvantage is associated with about a **1.7 percentage point decrease in graduation rate** and a **2.5 percentage point decrease in college enrollment**, and both effects are statistically significant at conventional levels. The negative effect on college enrollment is larger in magnitude, suggesting that disadvantage is especially important at the step from graduation to college.

For the **transition gap**, the estimated coefficient is positive—students in more disadvantaged neighborhoods appear somewhat more likely to "fall off" between graduation and college enrollment—but this effect is **not statistically significant** in our baseline model. We suspect this is partly because the model is deliberately simple (we do not include additional school-level controls or non-linear terms) and partly because the sample size is modest (128 high schools). We view this coefficient as a suggestive trend rather than strong causal evidence.







Coding Challenges

Highly correlated ACS variables. The poverty, income, unemployment, family structure, and rent-burden variables were strongly correlated, which made it hard to include them separately in regressions. We used **PCA** to extract a single Neighborhood Disadvantage Index that captures their common variation while avoiding multicollinearity. Additionally, our cross-sectional design cannot establish causality — the observed associations may reflect unmeasured confounders such as school funding or teacher quality. With only 128 schools in our sample, statistical power is also limited, which likely explains why the transition gap model did not reach significance despite a directionally consistent coefficient.

Streamlit App

Our `app.py` file turns the analysis into an interactive dashboard. Users can view the NDI choropleth and overlay schools, hover over points to see school names and outcomesand, and compare how outcomes change across the NDI distribution. This directly supports our research question by allowing users to explore how neighborhood context and school outcomes are related in different parts of Chicago.
The combined quantitative, spatial, and interactive approach gives a clear and policy-relevant picture: in Chicago, where a school is located is closely tied to whether its students make the leap from high school to college.